Housing Price Prediction Using Machine Learning

Ryan Maxam

Bellevue University – DSC 630

Professor Hua

August 23, 2026

Introduction of Topic/Problem

My objective for this project is to create a predictive analytics model that estimates the price of a residential home based on the property's characteristics. Accurate home price prediction is essential to the real estate industry because all parties, such as buyers, sellers, investors, and real estate professionals, rely on property valuations to make financial decisions. Traditional appraisal methods tend to be time-consuming and introduce bias, whereas predictive analytics can provide fast, data-driven estimates based on historical housing data.

For this project, I will use housing price datasets from Kaggle. The dataset contains a wealth of information about properties, including square footage, number of bedrooms, number of bathrooms, lot size, year built, and other characteristics. The target variable we will examine is the sale price of each property. In doing so, I will be able to develop a machine learning model to predict housing values for new properties.

Overview of Data Used

As mentioned, the dataset selected for this project is a publicly available housing price dataset available on Kaggle. Kaggle provides a vast number of well-documented datasets commonly used for predictive analytics and machine learning projects. The dataset I selected for my project contains historical housing sales records and various property attributes that could influence housing prices. (Yadav, 2026).

I chose this dataset because it contains both numerical and categorical variables, making it suitable for the necessary steps in machine learning, such as data preparation, exploratory data analysis, feature engineering, predictive modeling, and model evaluation. Additionally, housing prices are influenced by many factors, creating an opportunity to explore

which of these variables have the strongest impact on property values. The combination of numerical and categorical variables was chosen to compare interpretable regression methods with tree-based machine learning models.

When identifying the missing values, we found 99 missing cells across three variables. Area_SqFt contains 33 missing values, Rooms has 33 missing values, and Furnishing has 33 missing values. All of the other variables within the dataset are complete and don't contain any missing values.

Based on these missing value findings, I keep all of the records that have missing values because it would be unnecessary to reduce the available sample. For the missing numerical values we will be using the median for Area and rooms and the most frequent for the missing Furnishing values.

Descriptive Statistics

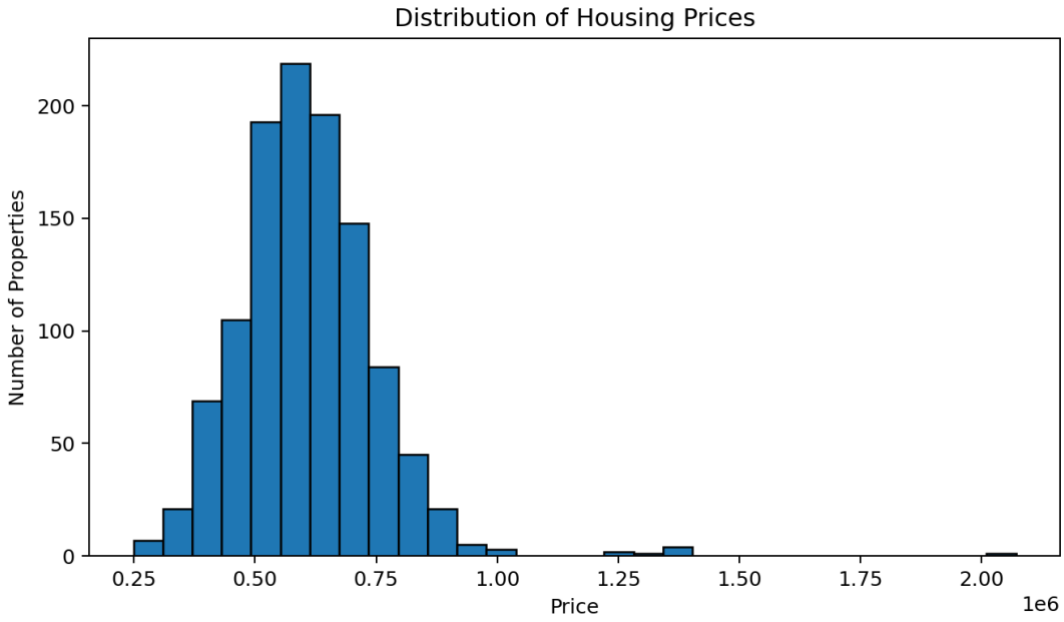
We found that the average property in the dataset is 2,237 square feet and has 4.57 rooms and was built in 2005. The mean housing price is $608,202, and the median is $602,541. The price ranges in this dataset are from the mid 200's up to 2 million. Based on the difference we have found in the mean and median, while also considering the Price skewness of 1.75, we can say that a right skewed target distribution has influenced the smaller number of high-priced properties.

Table 1

Descriptive Statistics

Variable	Mean	Median	Std. Dev.	Minimum	Maximum
Area_SqFt	2,237.48	2,201.55	731.10	700.00	10,267.12
Rooms	4.57	5.00	1.68	2.00	7.00
Build_Year	2,005.08	2,005.00	11.71	1,985.00	2,024.00
Price	608,201.51	602,541.00	143,195.81	248,640.00	2,071,401.84

Figure 1
Distribution of Housing Prices



Methods of Analysis

Correlation and Relationship Analysis

The strongest relationship with the Price is the square footage with a Pearson correlation of 0.862. On the other hand, Rooms has a weaker positive correlation of only 0.212 and the year the home was build has an even smaller correlation of 0.112. From these findings, we can see a direct correlation that the size of a property is very likely to be the most important factor in this dataset while the other features could have nonlinear effects.

Figure 3

Area vs. Housing Price



The Area_SqFt scatterplot shows us a clear positive relationship that larger properties are typically having higher prices. These patterns give us strong evidence to use linear regression as a useful baseline, but the spread of prices and large properties are suggesting that a tree based model could capture relationships that a linear equation does not.

Data Preparation Process

In the milestone 3 exploratory analysis I was able to find the specific preparation steps that were needed before we started diving into the development. The target variable in this analysis was Price, and some of the predictors contained: Area_SqFt, Rooms, Build_Year, Location, and Property_Type. The dataset was split up into different predictors and target values and used an 80/20 train test split that was used as a random state of 42. The test data were kept out during the model fitting in order to keep the final performance situated on observations that the model has not seen.

Missing numerical values in Area_SqFt and Rooms were imputed with the median. Missing Furnishing values were imputed with the most frequent category. Numerical predictors were standardized for the shared modeling pipeline, and categorical variables were converted to indicator variables with one-hot encoding. All imputation, scaling, and encoding steps were placed inside the modeling pipeline, so they were learned from the training data rather than from the full dataset, reducing the risk of data leakage.

The preliminary EDA also identified potential outliers in Area_SqFt and Price. These observations were retained for the first modeling pass because there was no evidence that they were data-entry errors. Retaining them allowed the models to be evaluated on the full observed price range. Their effect was monitored through RMSE, MAE, the actual-versus-predicted plot, and the residual plot.

Building Models

I created three regression models so that the project could compare an interpretable baseline with nonlinear tree-based approaches. Linear Regression was fit first. Decision Tree Regression was then fit with a maximum depth of 8 and a minimum of 5 observations. Finally, I used a Random Forest Regression model that was built using 300 trees and a minimum of 2 observations per leaf. The same train-test split, and preprocessing framework were also used for all three models so I could ensure the performance comparison was consistent.

Evaluation Methods

I evaluated the models using RMSE, MAE, R-squared, and five-fold cross-validation RMSE. Lower RMSE and MAE values indicate smaller prediction errors, while a higher R-squared value indicates that a larger proportion of the variation in housing prices is explained by the model.

Results & Findings Explained

Model Evaluation Results

Table 2

Model Evaluation Results

Model	Test RMSE	Test MAE	R-squared	5-Fold CV RMSE	CV SD
Linear Regression	34,259	20,084	0.944	34,526	6,436
Decision Tree Regression	62,236	49,804	0.817	66,658	6,325
Random Forest Regression	44,439	35,089	0.907	52,412	7,451

Figure 5

Model Error Comparison on Test Data



Model Performance

Once I completed the modeling, Linear Regression produced the strongest test-set performance. Its RMSE was approximately 34,259, its MAE was approximately 20,084, and its R-squared value was 0.944. Some of the inferences I made is that we can see that the model explained about 94.4% of the variation in housing prices in the held-out test set. The MAE

indicates that a typical absolute prediction error was about 20,084 price units. Because RMSE is higher than MAE, some observations produced substantially larger errors, which is consistent with the high-priced properties identified during Milestone 3.

Next, the Random Forest Regression was the second-best model, with an RMSE of approximately 44,439, an MAE of approximately 35,089, and an R-squared of 0.907. Decision Tree Regression performed worst of the three, with an RMSE of approximately 62,236 and an R-squared of 0.817. The single decision tree is to lose predictive accuracy when compared with both the linear model and the ensemble model.

The five-fold cross-validation results support the same overall conclusion. Linear Regression had the lowest average cross-validation RMSE at approximately 34,526, compared with 52,412 for Random Forest and 66,658 for Decision Tree Regression.

Figure 6

Linear Regression: Actual vs. Predicted Price

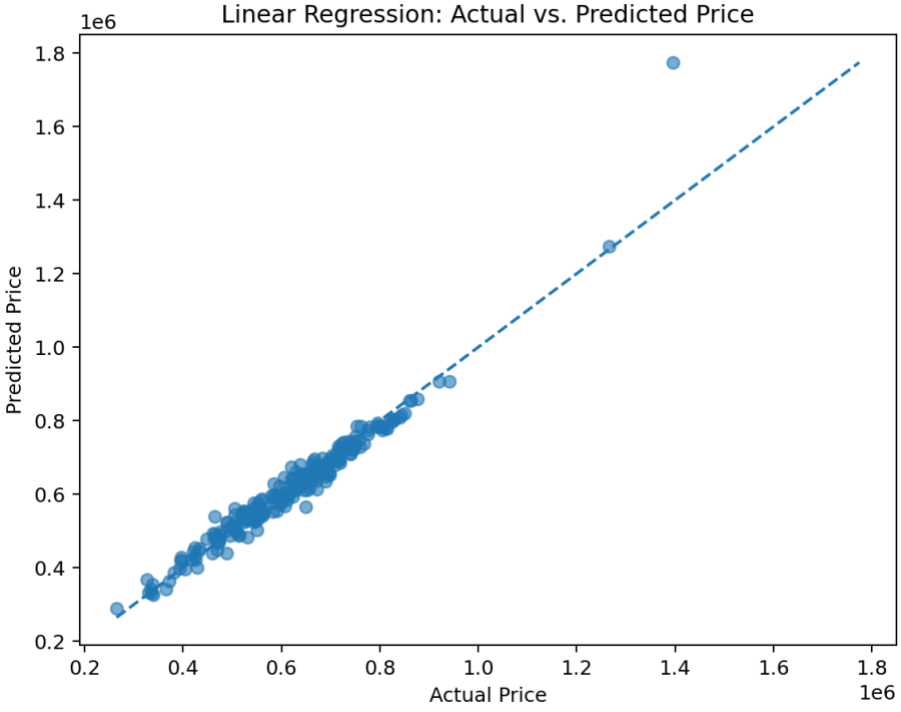
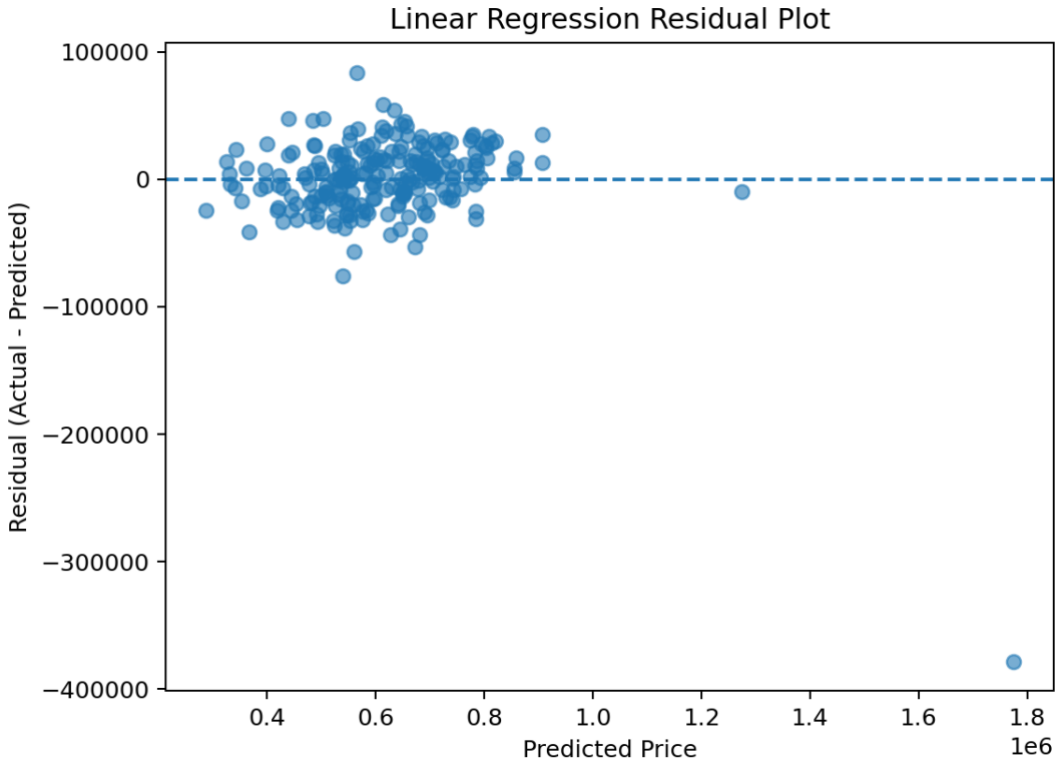


Figure 7
Linear Regression Residual Plot



Interpretation of Predictors

I found that Area_SqFt had the strongest numerical correlation with Price, and the fitted Linear Regression model continues to show that property size is an important driver after the other predictors are included. Because the categorical variables were one-hot encoded and the numerical variables were standardized, the model can also estimate price differences associated with property type, location, furnishing, street type, and pool status while holding the other variables constant. The coefficients are needing to be interpreted as associations within this dataset rather than as proof that a characteristic causes a change in housing price.

Conclusion

The technical results support the conclusion that the selected dataset can be used to predict housing prices with a high level of accuracy. Of the three models tested, Linear Regression is currently the preferred model because it produced the lowest test RMSE and MAE, the highest R-squared value, and the lowest average cross-validation RMSE. This result differs from the original expectation that Random Forest Regression would necessarily perform best. The comparison demonstrates why final model selection should be based on observed performance rather than model complexity.

For the final project, I recommend retaining Linear Regression as the primary model and using Random Forest as a secondary benchmark. The final analysis should continue to report RMSE, MAE, R-squared, and cross-validation results, because each metric provides a different view of model quality. I also recommend clearly communicating that predictions for unusually expensive properties may be less precise than predictions near the center of the market.

From a practical perspective, the model is best positioned as a decision-support tool that can provide an initial data-driven housing price estimate. It should not replace a professional appraisal or market analysis because the dataset may not contain every factor that influences a real property's value. Additional data such as more detailed geographic information, property condition, renovations, school districts, and current market conditions could improve future versions of the model.

References

Yadav, R. (2026). *House price prediction* [Dataset]. https://www.kaggle.com/datasets/raviiyadavv/house-price-predecition